

기초전기실험

5주차

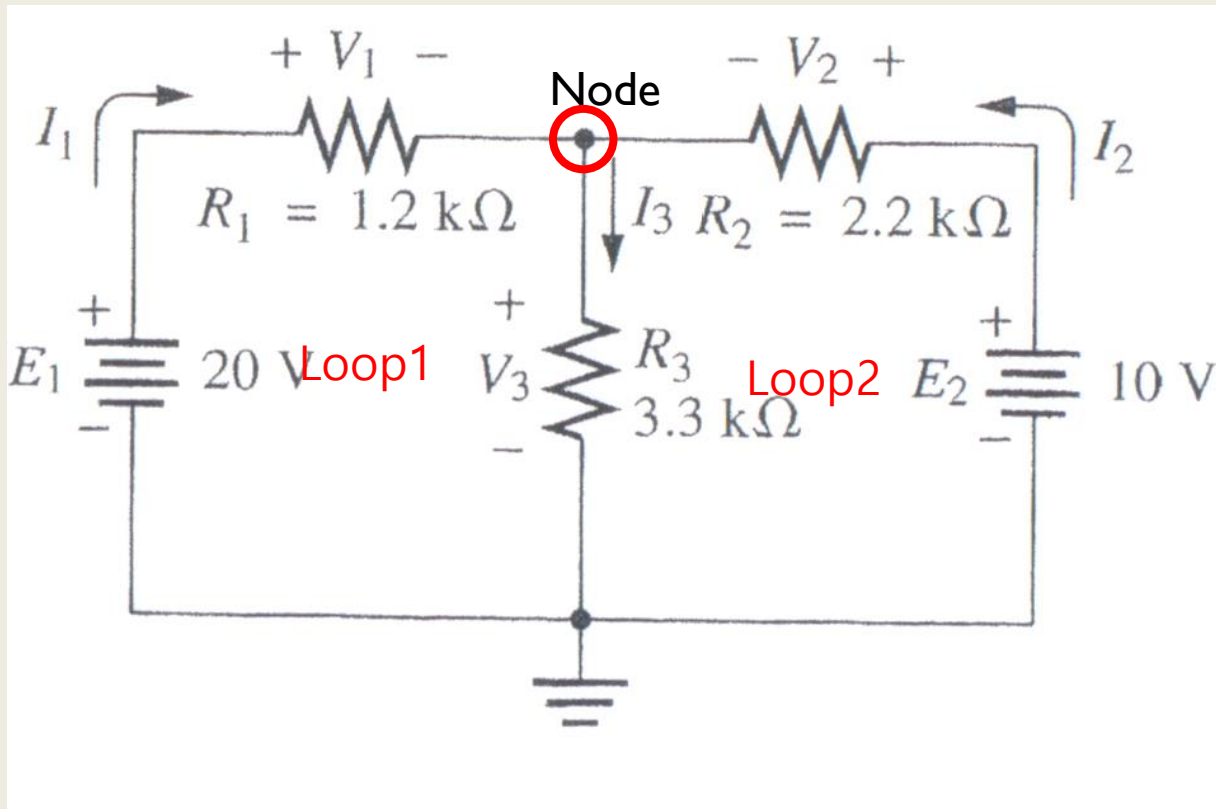
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5 주차 실험 목표

▣ DC Ch. 14 : Methods of Analysis

- ▣ 다양한 방법으로 회로를 분석해 보고, 측정을 통하여 이론을 확인한다.

BRANCH-CURRENT ANALYSIS



- ① 모든 Branch에 흐르는 전류의 방향 결정
- ② Node에 KCL 적용
- ③ Loop에 KVL 적용

BRANCH-CURRENT ANALYSIS

$$\textcircled{1} I_1 + I_2 = I_3$$

$$\textcircled{2} -20 + I_1 R_1 + I_3 R_3 = 0 \text{ (KVL in } L_1)$$

$$\textcircled{3} -10 + I_2 R_2 + I_3 R_3 = 0 \text{ (KVL in } L_2)$$

BRANCH-CURRENT ANALYSIS

② or ① choice

$$1.2kI_1 + 3.3kI_3 = 20 \Rightarrow 4.5kI_1 + 3.3kI_2 = 20$$

③ or ① choice

$$2.2kI_2 + 3.3kI_3 = 10 \Rightarrow 3.3kI_1 + 5.5kI_2 = 10$$

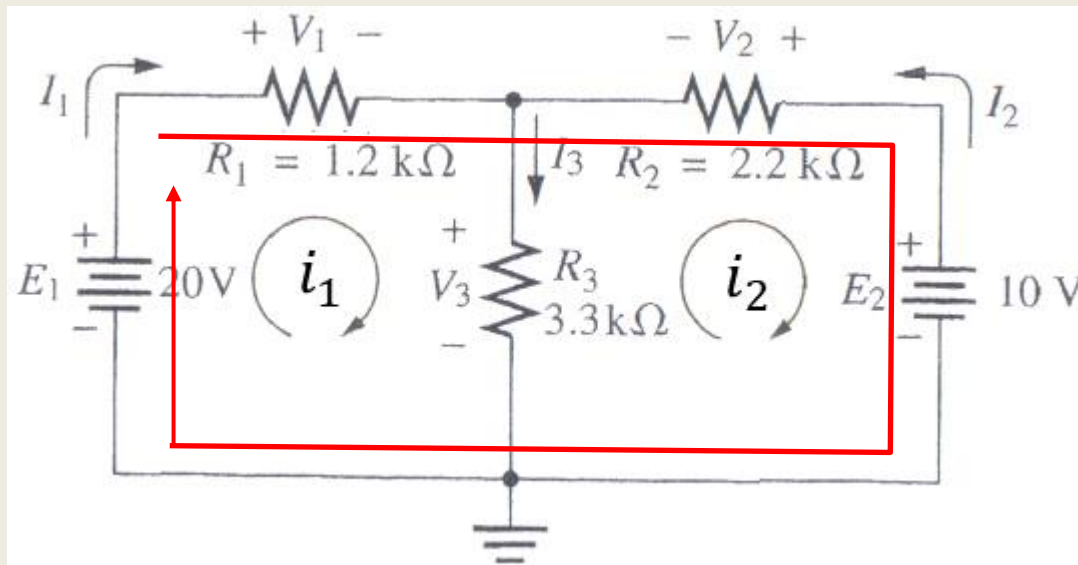
$$\begin{bmatrix} 4.5 & 3.3 \\ 3.3 & 5.5 \end{bmatrix} \begin{bmatrix} I_1 \\ I_2 \end{bmatrix} = \begin{bmatrix} 20 \\ 10 \end{bmatrix} \Rightarrow \begin{aligned} I_1 &= 5.5556 \text{ [mA]} \\ I_2 &= -1.5152 \text{ [mA]} \\ I_3 &= 4.0404 \text{ [mA]} \end{aligned}$$

$$\Rightarrow V_1 = 6.67 \text{ [V]}$$

$$V_2 = -3.33 \text{ [V]}$$

$$V_3 = 13.33 \text{ [V]}$$

MESH-CURRENT ANALYSIS



$$\begin{aligned} V_1 &= R_1 i_1 \\ V_2 &= R_2 i_2 \\ V_3 &= R_3 (i_1 - i_2) \\ I_1 &= i_1 \\ I_2 &= i_2 \\ I_3 &= i_1 - i_2 \end{aligned}$$

- ① 계산에 사용할 Loop 선정
- ② 선정한 Loop에 대해 KVL 적용

MESH-CURRENT ANALYSIS

① KVL in M_1

$$-E_1 + V_1 + V_3 + E_2 = 0$$

$$-20 + R_1 i_1 + R_3 (i_1 - i_2) + 10 = 0$$

$$(R_1 + R_3) i_1 - R_3 i_2 = 10$$

$$4.5k i_1 - 3.3k i_2 = 10$$

② KVL in M_2

$$-E_2 - V_3 + V_2 = 0$$

$$-10 - R_3 (i_1 - i_2) + R_2 i_2 = 0$$

$$-R_3 i_1 + (R_2 + R_3) i_2 = 10$$

$$-3.3k i_1 + 5.5k i_2 = 10$$

$$\begin{bmatrix} 4.5 & -3.3 \\ -3.3 & 5.5 \end{bmatrix} \begin{bmatrix} i_1 \\ i_2 \end{bmatrix} = \begin{bmatrix} 10 \\ 10 \end{bmatrix} \Rightarrow \begin{aligned} i_1 &= 6.3492 \text{ mA} \\ i_2 &= 5.6277 \text{ mA} \end{aligned}$$

$$\Rightarrow I_1 = 6.3492 \text{ [mA]}$$

$$I_2 = 5.6277 \text{ [mA]}$$

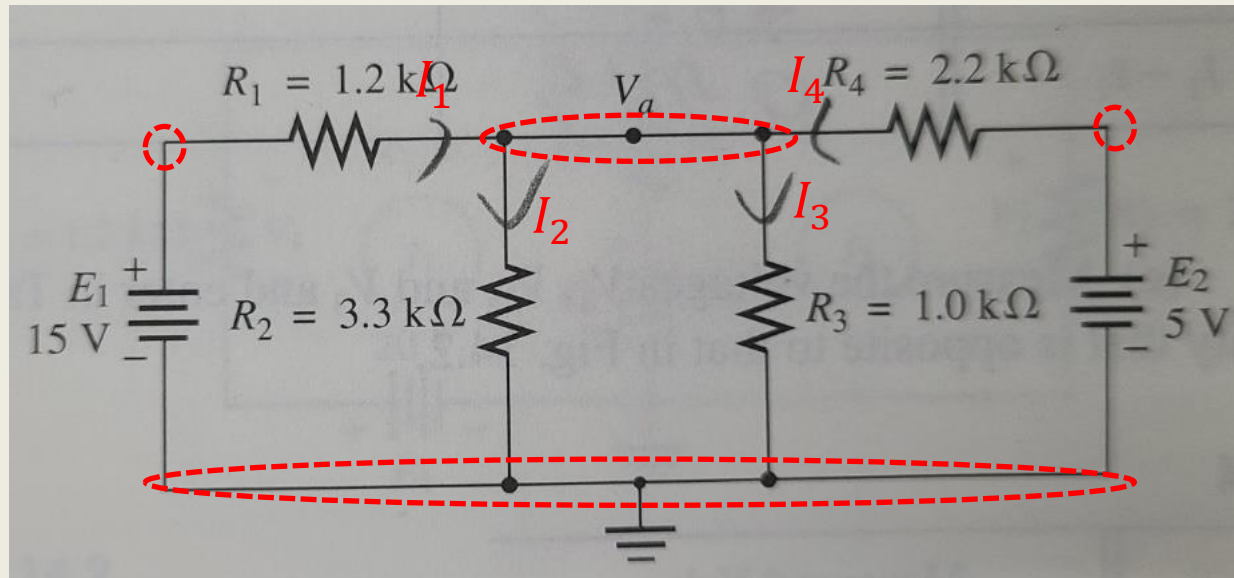
$$I_3 = 0.7215 \text{ [mA]}$$

$$V_1 = 7.619 \text{ [V]}$$

$$V_2 = 12.381 \text{ [V]}$$

$$V_3 = 2.381 \text{ [V]}$$

NODAL ANALYSIS



○: Node

- ① 계산에 사용할 Node 선정
- ② 선정한 Node에 대해 KCL 적용

NODAL ANALYSIS

$$I_1 - I_2 - I_3 + I_4 = 0$$

$$I_1 = \frac{E_1 - V_a}{R_1}$$

$$I_3 = \frac{V_a}{R_3}$$

$$I_2 = \frac{V_a}{R_2}$$

$$I_4 = \frac{E_2 - V_a}{R_4}$$

NODAL ANALYSIS

$$\frac{15 - V_a}{1.2k} - \frac{V_a}{3.3k} - \frac{V_a}{1k} + \frac{5 - V_a}{2.2k} = 0$$

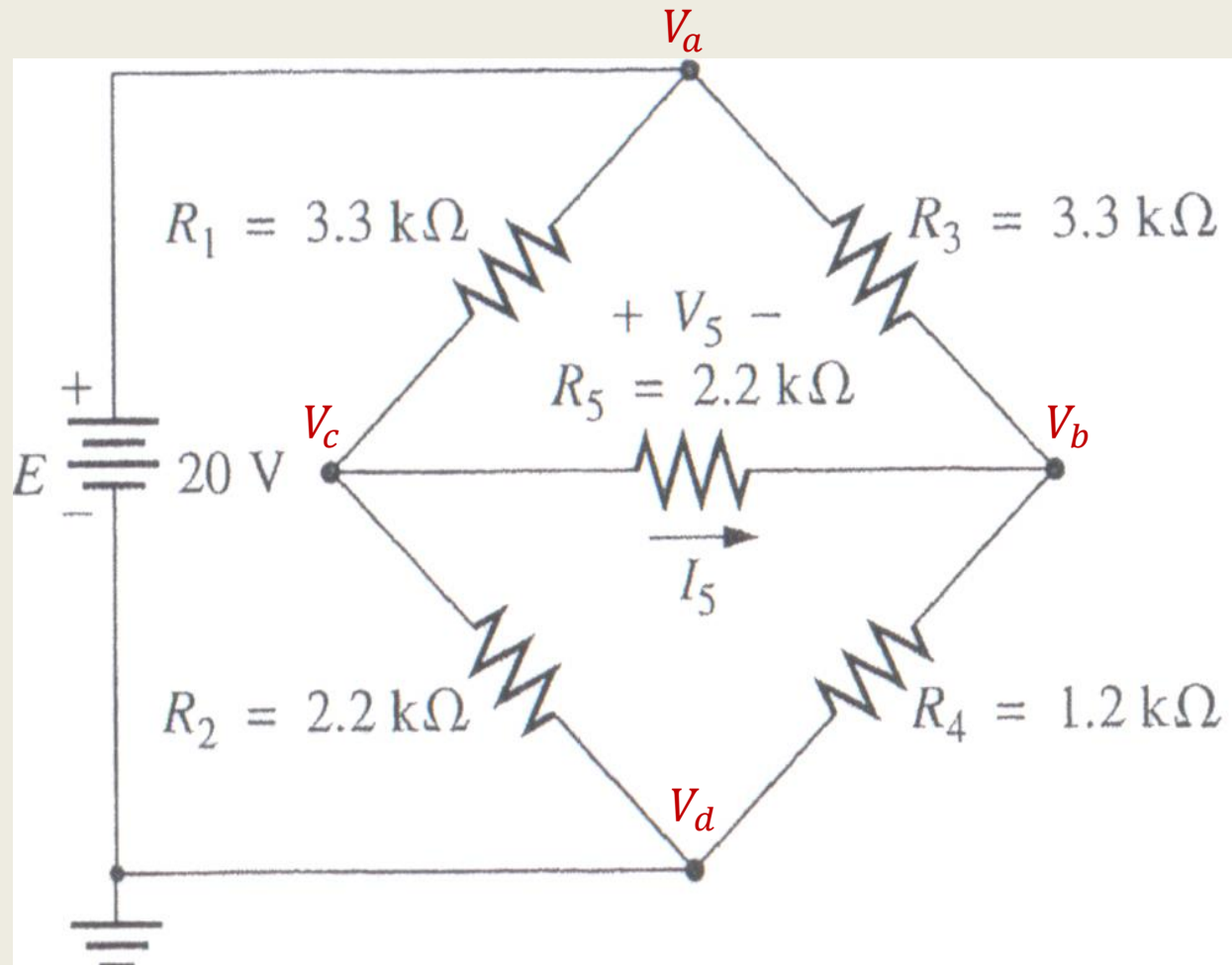
$$V_a \left(-\frac{1}{1.2k} - \frac{1}{3.3k} - \frac{1}{1k} - \frac{1}{2.2k} \right) + \frac{15}{1.2k} + \frac{5}{2.2k} = 0$$

$$V_a = \frac{\frac{15}{1.2} + \frac{5}{2.2}}{\frac{1}{1.2} + \frac{1}{3.3} + 1 + \frac{1}{2.2}} = 5.7017$$

$$\therefore I_{R_1} = I_1 = \frac{15 - 5.7017}{1.2k} = 7.7486 \text{ [mA]}$$

$$I_{R_3} = I_3 = \frac{5.7017}{1k} = 5.7017 \text{ [mA]}$$

BRIDGE NETWORK



BRIDGE NETWORK

$$V_a = 20, \quad V_d = 0$$

node b.

$$\frac{V_b - V_c}{R_5} + \frac{V_b - V_a}{R_3} + \frac{V_b - V_d}{R_4} = 0$$

$$\begin{aligned} \frac{V_b - V_c}{2.2k} + \frac{V_b - 20}{3.3k} + \frac{V_b}{1.2k} &= \left(\frac{1}{2.2k} + \frac{1}{3.3k} + \frac{1}{1.2k} \right) V_b \\ &\quad - \frac{1}{2.2k} V_c - \frac{20}{3.3k} = 0 \\ \frac{35}{22} V_b - \frac{5}{11} V_c &= \frac{200}{33} \end{aligned}$$

BRIDGE NETWORK

node C

$$\frac{35}{22}V_b - \frac{5}{11}V_c = \frac{200}{33}$$

~~2.2k~~ ~~3.3k~~

$$\frac{V_c - V_b}{R_5} + \frac{V_c - V_a}{R_1} + \frac{V_c - V_d}{R_2} = 0$$

$$\frac{V_c - V_b}{2.2k} + \frac{V_c - 20}{3.3k} + \frac{V_c - 0}{2.2k} = \left(\frac{1}{2.2k} + \frac{1}{3.3k} + \frac{1}{2.2k} \right) V_c - \frac{1}{2.2k} V_b - \frac{20}{3.3k} = 0$$

$$-\frac{5}{11}V_b + \frac{40}{33}V_c = \frac{200}{33}$$

$$\begin{bmatrix} 1.5909 & -0.4545 \\ -0.4545 & 1.2121 \end{bmatrix} \begin{bmatrix} V_b \\ V_c \end{bmatrix} = \begin{bmatrix} 6.0606 \\ 6.0606 \end{bmatrix}$$

$$V_b = 5.8667, \quad V_c = 7.2000$$

$$\therefore V_s = V_c - V_b = 1.33 \text{ [V]}$$

$$I_s = \frac{1.33}{2.2k} = 0.6045 \text{ [mA]}$$

5 주차 실험

▣ DC Ch. 14 : Methods of Analysis

- ▣ part1($1.2\text{k}\Omega \times 1$, $2.2\text{k}\Omega \times 1$, $3.3\text{k}\Omega \times 1$)
 - ▣ FIG 14.1 구성
 - ▣ 전류 측정하여 Table 14.1 완성(전류 방향 주의)
 - ▣ 전압 측정하여 Table 14.2 완성
- ▣ part2($1.2\text{k}\Omega \times 1$, $2.2\text{k}\Omega \times 1$, $3.3\text{k}\Omega \times 1$)
 - ▣ FIG 14.2 구성
 - ▣ 전류 측정하여 Table 14.3 완성(전류 방향 주의)
 - ▣ 전압 측정하여 Table 14.4 완성
- ▣ part3($1.2\text{k}\Omega \times 1$, $2.2\text{k}\Omega \times 1$, $3.3\text{k}\Omega \times 1$)
 - ▣ FIG 14.3 구성
 - ▣ 전압 측정하여 Table 14.5 완성
 - ▣ 전류 측정하여 Table 14.6 완성 (전류 방향 표시할 것 ex) $7.7\text{mA}(\rightarrow)$)
- ▣ Part4($1.2\text{k}\Omega \times 1$, $2.2\text{k}\Omega \times 2$, $3.3\text{k}\Omega \times 2$)
 - ▣ FIG 14.4 구성
 - ▣ 전압 및 전류 측정하여 Table 14.7 완성(전류 방향 주의)